

BRAC University

Department of Computer Science and Engineering

CSE111: Programming Language II

Examination: Quiz #1

Semester: Summer 2022

Date: ____ / ____ / 2022

Time: 40 Minutes

ID: _____	Name: (Please write in CAPITAL LETTERS)	Section: 42
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- Read questions carefully.
- Understanding the question is part of the exam, please do not ask questions.

1. Write a simple encryption function, `usis_encryption(...)`, in Python. This function should be able to return an encrypted password from the given password as `str`. The process is divided into two parts. [CO1]

First, according to the ASCII table:

(9 Marks)

- Each digit must be shifted ($\text{LAST_DIGIT_OF_ID} \% 2$) positions to the right.
- Each lowercase letter must be shifted to ($\text{LAST_DIGIT_OF_ID} \% 2 + 2$) positions to the right.
- Each uppercase letter must be shifted to ($\text{LAST_DIGIT_OF_ID} \% 3 + 5$) positions to the right.

After shifting the positions:

(4 Marks)

- The encrypted password should be divided into two parts.
- The second half should be reversed and finally concat the two parts and return the encrypted password.

```
# Your code here
```

```
usis_password = input("Please enter your usis password: ")
usis_password = usis_encryption(usis_password)
print(usis_password)
```

For ID: **18341159**

Input Sample	Output Sample
abcABC123	defF432HG
Summer2022	Xxp-ph3313u

2. Answer any of the two questions below, both for bonus. (7 Marks - each)

Note: Bonus will be given if at least one question is answered correctly. Otherwise, the highest number will be considered and no bonus will be given.

- a. Given a tuple - your program does not need to take input - write a function, `tuple_sorter(...)`, in Python that will return a sorted tuple in descending order. **[CO1]**

You are not allowed to use:

- ✗ `list()` - built in function
- ✗ `sort()`, `sorted()` - built in function

Run Sample	Output Sample
<code>print(tuple_sorter((1, 8, 3, 4, 1, 1, 5, 9)))</code>	<code>(9, 8, 5, 4, 3, 1, 1, 1)</code>
<code>print(tuple_sorter((3, 0, 1, 0, 1, 4, 3, 1)))</code>	<code>(4, 3, 3, 1, 1, 1, 0, 0)</code>

- b. Write a python program that calculates the sum of **N** given lists and prints the second highest sum and its respective list. Input starts with **N** and followed by **N** lists. **[CO1]**

Input Sample	Output Sample
4 1 2 3 4 5 6 10 11 12 7 8 9	24 [7, 8, 9]